Exercise 2.3-2.4

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2.3.1. 设 μ 是 X 上的一个外测度.

(1) 设 $Z \subset X$ 且 $\mu(Z) = 0$. 证明对任意 $E \subset X$ 有

$$\mu(E \cup Z) = \mu(E \setminus Z) = \mu(E).$$

(换言之, 对 E 加上或减去一个零测集, 不改变 E 的外测度.)

证明.

$$\mu(E) \leq \mu(E \cup Z) \leq \mu(E) + \mu(Z) = \mu(E) \Longrightarrow \mu(E \cup Z) = \mu(E)$$

$$\mu(E) \geq \mu(E \setminus Z) \geq \mu(E) - \mu(E \cap Z) \geq \mu(E) \Longrightarrow \mu(E \setminus Z) = \mu(E)$$

□

(2) 设 $A, B \subset X$ 且 $\mu(A) < \infty, \mu(B) < \infty$, 证明

$$|\mu(A) - \mu(B)| \leq \mu(A \Delta B)$$

这里和下面,

$$A \Delta B = (A \setminus B) \cup (B \setminus A),$$

称之为 A 与 B 的对称差.

证明.
$$\mu(A)=\mu((A\setminus B)\cup(A\cap B))\le\mu(A\setminus B)+\mu(A\cap B)\le\mu(A\Delta B)+\mu(B)$$
$$\mu(B)=\mu((B\setminus A)\cup(A\cap B))\le\mu(B\setminus A)+\mu(A\cap B)\le\mu(A\Delta B)+\mu(A)$$
Then
$$|\mu(A)-\mu(B)|=\max\{\mu(A)-\mu(B),\mu(B)-\mu(A)\}\le\mu(A\Delta B)$$
□

(3) 设 $A,B,C\subset X$ 且有 $\mu(A\Delta B)=0,\mu(B\Delta C)=0$. 证明 $\mu(A\Delta C)=0$.

证明. We first prove that
$$A\Delta C\subset A\Delta B\cup B\Delta C$$
$\forall x\in A\Delta C$, without loss of generality, we assume that $x\in A\setminus C$. If $x\in B$, then $x\in B\setminus C\subset B\Delta C$; if $x\notin B$, then $x\in A\setminus B\subset A\Delta B$. Therefore,
$$0\le\mu(A\Delta C)\le\mu(A\Delta B)+\mu(B\Delta C)=0\Longrightarrow\mu(A\Delta C)=0$$
□

2.3.2. 设 μ 是集合 X 上的一个外测度, $(X,\mathcal{M},\mu)$ 是由此导出的测度空间.

(1) 设 $A\subset B\subset X$. 假设 A 是 $\mathcal{M}$-可测集 (即 $A\in\mathcal{M}$) 且 $\mu(A)=\mu(B)<\infty$. 证明 B 也是 $\mathcal{M}$-可测集.

证明. $\forall T\subset X$, we have
$$\mu(T)=\mu(A\cap T)+\mu(A^c\cap T)$$
Since $A\subset B$, $A\in\mathcal{M}$, then $\mu(B)=\mu(A\cap B)+\mu(B\cap A^c)$. Hence
$$\mu(B\setminus A)=\mu(B)-\mu(A\cap B)=\mu(B)-\mu(A)=0$$

Let $C = B \setminus A$, then

$$\mu(B \cap T) + \mu(B^c \cap T) = \mu(B \cap T \setminus C) + \mu(B^c \cap T \cup C) = \mu(A \cap T) + \mu(A^c \cap T) = \mu(T)$$

This establishes that B is a measurable set. $\square$

(2) 设 $A, B \subset X$ 都是 $\mathcal{M}$-可测集. 证明 $\mu(A \cap B) + \mu(A \cup B) = \mu(A) + \mu(B)$.

证明.

$$\begin{aligned}
\mu(A \cap B) + \mu(A \cup B) &= \mu(A \cap B) + \mu((A \cup B) \cap A) + \mu((A \cup B) \cap A^c) \\
&= \mu(A \cap B) + \mu(A) + \mu(B \setminus A) \\
&= \mu(A \cap B) + \mu(A) + \mu(B) - \mu(A \cap B) \\
&= \mu(A) + \mu(B)
\end{aligned}$$

$\square$

2.3.3. 设 μ 是集合 X 上的一个外测度, $(X, \mathcal{M}, \mu)$ 是由此导出的测度空间.

设 $\varphi : X \to X$ 是一个单满射且保持集合的外测度不变:

$$\mu(\varphi(E)) = c\mu(E) \qquad \forall E \subset X.$$

其中 $0 < c < \infty$ 为常数. 证明 φ 把 $\mathcal{M}$-可测集映为 $\mathcal{M}$-可测集.

证明. $\forall A \in \mathcal{M}$, $\forall T \subset X$, we next show that

$$\mu(T) = \mu(\varphi(A) \cap T) + \mu(\varphi(A)^c \cap T)$$

Since φ is bijective, $\exists S \subset X$, s.t. $T = \varphi(S)$. Therefore,

$$\begin{aligned}
&\mu(\varphi(A) \cap T) + \mu(\varphi(A)^c \cap T) = \mu(\varphi(A) \cap \varphi(S)) + \mu(\varphi(A)^c \cap \varphi(S)) \\
=&\mu(\varphi(A \cap S)) + \mu(\varphi(A^c \cap S)) = c(\mu(A \cap S) + \mu(A^c \cap S)) \\
=&c\mu(S) = \mu(\varphi(S)) = \mu(T)
\end{aligned}$$

$\square$

2.3.4*. 设 $(X, \mathcal{M}, \mu)$ 是由 X 上的外测度 μ 导出的测度空间, 它具有等测包性性质:

对任意 $A \subset X$, 存在 $H_A \in \mathcal{M}, H_A \supset A$, 使得 $\mu(H_A) = \mu(A)$.

(a) 设 $A, B \subset X$ 满足

$$A \cup B \in \mathcal{M}, \quad \mu(A \cup B) = \mu(A) + \mu(B) < \infty. \tag{1}$$

证明 $A, B \in \mathcal{M}$ 且 $\mu(A \cap B) = 0$.

证明. By theexistence of a measurable envelope with equal measure, we have $H_A \supset A, H_B \supset B \in \mathcal{M}$ s.t. $\mu(H_A) = \mu(A), \mu(H_B) = \mu(B) \Longrightarrow A \cup B \subset H_A \cup H_B$. Then

$$\mu(A) + \mu(B) = \mu(A \cup B) \le \mu(H_A \cup H_B) \le \mu(H_A) + \mu(H_B) = \mu(A) + \mu(B).$$

This gives

$$\mu(H_A \cup H_B) = \mu(H_A) + \mu(H_B).$$

Since $H_A, H_B \in \mathcal{M}$,

$$\mu(H_A \cup H_B) = \mu(H_A) + \mu(H_B) - \mu(H_A \cap H_B).$$

Hence

$$\mu(H_A \cap H_B) = 0 \Longrightarrow \mu(A \cap B) \le \mu(H_A \cap H_B) = 0.$$

On the other hand, note that

$$A \cup B \in \mathcal{M}, \qquad A \cup B \subset H_A \cup H_B, \qquad \mu(A \cup B) = \mu(H_A \cup H_B) < \infty,$$

By 2.3.2(1)

$$H_A \cup H_B \in \mathcal{M} \qquad \mu\big((H_A \cup H_B) \setminus (A \cup B)\big) = 0$$

We next show that $\mu(H_A \setminus A) = 0$. For $x \in H_A \setminus A$, if $x \in B$, then $x \in H_A \cap H_B$; if $x \notin B$, then $x \in (H_A \cup H_B) \setminus (A \cup B)$. Therefore,

$$H_A \setminus A \subset (H_A \cap H_B) \cup \big((H_A \cup H_B) \setminus (A \cup B)\big).$$

Then

$$\mu(H_A \setminus A) \le \mu(H_A \cap H_B) + \mu\big((H_A \cup H_B) \setminus (A \cup B)\big) = 0.$$

Similarly,

$$\mu(H_B \setminus B) = 0.$$

We first show that any zero measure set is measurable. If $\mu(N) = 0$, then for any $T \subset X$,

$$\mu(T) \le \mu(T \cap N) + \mu(T \cap N^c) \le \mu(N) + \mu(T) = \mu(T),$$

i.e.

$$\mu(T) = \mu(T \cap N) + \mu(T \cap N^c),$$

Therefoe $N \in \mathcal{M}$.

Since $H_A \setminus A$ is a zero measure set, $H_A \setminus A \in \mathcal{M}$. Together with $H_A \in \mathcal{M}$,

$$A = H_A \setminus (H_A \setminus A) \in \mathcal{M}.$$

Similarly $B \in \mathcal{M}$. □

(b) 设集合 $E_k \subset X$ $(k = 1, 2, \dots, N)$ 满足

$$\bigcup_{k=1}^{N} E_k \in \mathcal{M}, \quad \mu\left(\bigcup_{k=1}^{N} E_k\right) = \sum_{k=1}^{N} \mu(E_k) < \infty. \tag{2}$$

证明 $E_k \in \mathcal{M}, k = 1, 2, \dots, N$ 且 E_k 互不重叠, 即当 $i \neq j$ 时 $\mu(E_i \cap E_j) = 0$.

证明. We prove this by the induction of N. $N = 2$ is (a). We assume that the conclusion holds for $N - 1$. When it comes to N, let

$$F = \bigcup_{k=1}^{N-1} E_k \Longrightarrow \mu(F) \le \sum_{k=1}^{N-1} \mu(E_k).$$

Since

$$\bigcup_{k=1}^{N} E_k = F \cup E_N,$$

we have

$$\sum_{k=1}^{N} \mu(E_k) = \mu\left(\bigcup_{k=1}^{N} E_k\right) \le \mu(F) + \mu(E_N) \le \sum_{k=1}^{N-1} \mu(E_k) + \mu(E_N) = \sum_{k=1}^{N} \mu(E_k).$$

This gives

$$\mu(F) = \sum_{k=1}^{N-1} \mu(E_k) \qquad \mu(F \cup E_N) = \mu(F) + \mu(E_N) < \infty.$$

Then by (a),

$$F \in \mathcal{M}, \qquad E_N \in \mathcal{M}, \qquad \mu(F \cap E_N) = 0.$$

Applying the induction hypothesis to $E_1, \ldots, E_{N-1}$, we get

$$E_k \in \mathcal{M} \quad (k = 1, \ldots, N-1), \qquad \mu(E_i \cap E_j) = 0 \quad (1 \le i < j \le N-1).$$

Also, $\forall 1 \le i \le N-1$,

$$E_i \cap E_N \subset F \cap E_N,$$

Hence,

$$\mu(E_i \cap E_N) = 0.$$

Therefore, $E_k \in \mathcal{M}$ $(k = 1, \ldots, N)$ and $\mu(E_i \cap E_j) = 0$ for any $1 \le i < j \le N$. □

(c) 设集合 $E_k \subset X$ $(k = 1, 2, 3, \ldots)$ 满足

$$\bigcup_{k=1}^{\infty} E_k \in \mathcal{M}, \quad \mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k) < \infty. \tag{3}$$

证明 $E_k \in \mathcal{M}, k = 1, 2, 3, \ldots,$ 且 E_k 互不重叠.

证明. We write

$$U = \bigcup_{k=1}^{\infty} E_k, \qquad F_n = \bigcup_{k=1}^{n} E_k, \qquad G_n = \bigcup_{k=n+1}^{\infty} E_k.$$

then $U = F_n \cup G_n$.

We first show that $\forall n \in \mathbb{N}$, $\mu(F_n) = \sum_{k=1}^{n} \mu(E_k)$ and $\mu(G_n) = \sum_{k=n+1}^{\infty} \mu(E_k)$. First we have

$$\mu(F_n) \le \sum_{k=1}^{n} \mu(E_k) \qquad \mu(G_n) \le \sum_{k=n+1}^{\infty} \mu(E_k)$$

Then

$$\sum_{k=1}^{\infty} \mu(E_k) = \mu(U) \le \mu(F_n) + \mu(G_n) \le \sum_{k=1}^{n} \mu(E_k) + \sum_{k=n+1}^{\infty} \mu(E_k) = \sum_{k=1}^{\infty} \mu(E_k).$$

$$\Longrightarrow \mu(F_n) = \sum_{k=1}^{n} \mu(E_k), \qquad \mu(G_n) = \sum_{k=n+1}^{\infty} \mu(E_k), \qquad \mu(U) = \mu(F_n) + \mu(G_n)$$

Note that $U = F_n \cup G_n \in \mathcal{M}$, $\mu(U) < \infty$, applying (a) gives

$$F_n \in \mathcal{M}, \qquad G_n \in \mathcal{M}, \qquad \mu(F_n \cap G_n) = 0.$$

For fixed $n \in \mathbb{N}$, we have

$$F_n = \bigcup_{k=1}^{n} E_k \in \mathcal{M}, \qquad \mu(F_n) = \sum_{k=1}^{n} \mu(E_k) < \infty,$$

From (b) we have

$$E_1, \ldots, E_n \in \mathcal{M}, \qquad \mu(E_i \cap E_j) = 0 \quad (1 \le i < j \le n).$$

Then $E_k \in \mathcal{M}$, $\forall k \in \mathbb{N}$.
$\forall i \ne j$, let $n \ge \max\{i, j\}$, we have

$$\mu(E_i \cap E_j) = 0.$$

This gives $\{E_k\}_{k\ge 1}$ are mutually disjoint up to a zero measure set. □

2.3.5. 设 μ 是 $\mathbb{R}^d$ 上的一个外测度, $E \subset \mathbb{R}^d$.

假设对任意 $x \in E$, 存在 $\delta_x > 0$ 使得 $\mu(E \cap B(x, \delta_x)) = 0$. 证明 $\mu(E) = 0$.

证明. $\forall x \in E$, $\exists \delta = \delta_x > 0$ s.t. $\mu(E \cap B(x, \delta_x)) = 0$, then we have

$$E \subset \bigcup_{x \in E} B(x, \delta_x)$$

Since any open cover of E can be replaced by a countable subcover, there exists $\{x_i\}_{i\ge 1} \subset E$ s.t.

$$E \subset \bigcup_{x \in E} B(x, \delta_x) \subset \bigcup_{i \ge 1} B(x_i, \delta_{x_i})$$

Then

$$E = \bigcup_{i \ge 1} E \cap B(x_i, \delta_{x_i}) \Longrightarrow \mu(E) \le \sum_{i \ge 1} \mu(E \cap B(x_i, \delta_{x_i})) = 0$$

□

2.3.6. 设 $A, B_k \subset \mathbb{R}^d, k = 1, 2, \ldots, N$.

证明

$$\operatorname{dist}\left(A, \bigcup_{k=1}^{N} B_k\right) = \min_{1 \le k \le N} \operatorname{dist}(A, B_k).$$

证明. $\forall x \in A, y \in \bigcup_{k=1}^{N} B_k$, $\exists 1 \leq k \leq N$ s.t. $y \in B_k$, then

$$|x-y| \geq dist(x, B_k) \geq dist(A, B_k) \geq \min_{1\leq k\leq N} dist(A, B_k)$$

Taking the infimum of the left side gives

$$dist(A, \bigcup_{k=1}^{N} B_k) \geq \min_{1\leq k\leq N} dist(A, B_k)$$

On the other side, $\forall 1 \leq k \leq N$, $B_k \subset \bigcup_{k=1}^{N} B_k$, then

$$dist(A, \bigcup_{k=1}^{N} B_k) \leq dist(A, B_k) \Longrightarrow dist(A, \bigcup_{k=1}^{N} B_k) \leq \min_{1\leq k\leq N} dist(A, B_k)$$

Therefore,

$$dist(A, \bigcup_{k=1}^{N} B_k) = \min_{1\leq k\leq N} dist(A, B_k)$$

□

2.3.7(距离可加性). 设 μ 是 $\mathbb{R}^d$ 上的一个度量外测度.

设 $E_1, E_2, \ldots, E_N \subset \mathbb{R}^d$ 满足

$$\operatorname{dist}(E_i, E_j) > 0, \quad i \neq j, \quad i, j = 1, 2, \ldots, N.$$

证明

$$\mu\left(\bigcup_{k=1}^{N} E_k\right) = \sum_{k=1}^{N} \mu(E_k).$$

然后将这一结论推广到可数多个集合的情形: 设 $E_k \subset \mathbb{R}^d, k = 1, 2, 3, \ldots$ 满足

$$\operatorname{dist}(E_i, E_j) > 0, \quad i \neq j, \quad i, j = 1, 2, 3 \ldots.$$

证明

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

证明. We prove this be the inductio of N. $N=2$ holds as the definition. We assume that the equality holds for N. When it comes to $N+1$,

$$dist(E_{N+1},\bigcup_{k=1}^{N}E_k)\geq \min_{1\leq k\leq N} dist(E_{N+1},E_k)>0$$

Then

$$\mu(\bigcup_{k=1}^{N+1}E_k)=\mu(\bigcup_{k=1}^{N}E_k\cup E_{N+1})=\mu(\bigcup_{k=1}^{N}E_k)+\mu(E_{N+1})=\sum_{k=1}^{N}\mu(E_k)+\mu(E_{N+1})=\sum_{k=1}^{N+1}\mu(E_k)$$

Suppose $E_k\subset\mathcal{R}^d, k=1,2,3,\ldots$ satisfy

$$dist(E_i,E_j)>0,\quad i\neq j,\quad i,j=1,2,3,...$$

$\forall N\in\mathbb{N}$, we have

$$\sum_{k=1}^{\infty}\mu(E_k)\geq\mu(\bigcup_{k=1}^{\infty}E_k)\geq\mu(\bigcup_{k=1}^{N}E_k)=\sum_{k=1}^{N}\mu(E_k)\to\sum_{k=1}^{\infty}\mu(E_k)$$

This gives

$$\mu(\bigcup_{k=1}^{\infty}E_k)=\sum_{k=1}^{\infty}\mu(E_k)$$

□

2.4.1. 设 $(\mathbb{R}^d,\mathcal{M},\mu)$ 为一测度空间, 其中 σ-代数 $\mathcal{M}$ 包含 $\mathbb{R}^d$ 中的所有 Borel 集, μ 是 $\mathcal{M}$ 上的一个正则测度.

设 $E\subset\mathbb{R}^d$ 为一个有界集 (非空). 对任意 $\varepsilon>0$, 令

$$E_{<\varepsilon}=\{x\in\mathbb{R}^d\mid \mathrm{dist}(x,E)<\varepsilon\},\quad E_{\leq\varepsilon}=\{x\in\mathbb{R}^d\mid \mathrm{dist}(x,E)\leq\varepsilon\},\quad K=\overline{E}.$$

证明

$$(1)\quad K_{<\varepsilon}=E_{<\varepsilon},\quad K_{\leq\varepsilon}=E_{\leq\varepsilon}\qquad\forall\varepsilon>0,$$

$$(2)\quad \lim_{\varepsilon\to0^+}\mu(K_{\leq\varepsilon})=\lim_{\varepsilon\to0^+}\mu(K_{<\varepsilon})=\mu(K),$$

从而有

$$(3)\quad \lim_{\varepsilon\to0^+}\mu(K_{\leq\varepsilon}\setminus K)=\lim_{\varepsilon\to0^+}\mu(K_{<\varepsilon}\setminus K)=0.$$

First, since E is bounded, together with $K = \bar{E}$ is closed, K is compact, which gives $\mu(K) < \infty$, $\mu(K_{\leq 1}) < \infty$ by the regularity of μ.

(1)

证明. $\forall x \in K_{<\epsilon}$, since K is a closed set, $\exists y \in K$, $|x-y| = dist(x, K) < \epsilon$. Since $K = \bar{E}$, $\exists E \supset \{y_k\}_{k\geq 1} \to y$, $dist(x, E) \leq \inf_{k\geq 1} |x - y_k| < \epsilon \Longrightarrow x \in E_{<\epsilon}$.

On the other hand, $\forall x \in E_{<\epsilon}$,

$$K \supset E \Longrightarrow \inf_{y\in K} |x-y| \leq \inf_{y\in E} |x-y| \Longrightarrow dist(x, K) \leq dist(x, E) < \epsilon$$

Therefore, $K_{<\epsilon} = E_{<\epsilon}$.

$\forall x \in K_{\leq\epsilon}$, since K is a closed set, $\exists y \in K$, $|x-y| = dist(x, K) \leq \epsilon$. Since $K = \bar{E}$, $\exists E \supset \{y_k\}_{k\geq 1} \to y$, $dist(x, E) \leq \inf_{k\geq 1} |x - y_k| \leq \epsilon \Longrightarrow x \in E_{\leq\epsilon}$.

On the other hand, $\forall x \in E_{\leq\epsilon}$,

$$K \supset E \Longrightarrow \inf_{y\in K} |x-y| \leq \inf_{y\in E} |x-y| \Longrightarrow dist(x, K) \leq dist(x, E) \leq \epsilon$$

Therefore, $K_{\leq\epsilon} = E_{\leq\epsilon}$. □

(2)

证明. We know that $\forall \epsilon > \epsilon' > 0$, $K_{<\epsilon'} \subset K_{<\epsilon}$, then

$$\lim_{\epsilon\to 0^+} \mu(K_{<\epsilon}) = \mu(\bigcap_{\epsilon>0} K_{<\epsilon})$$

We next show that $\bigcap_{\epsilon>0} K_{<\epsilon} = K$. $\forall x \in \bigcap_{\epsilon>0} K_{<\epsilon}$, $\forall \epsilon > 0$, $dist(x, K) < \epsilon \Longrightarrow dist(x, K) = 0$. Since K is a closed set, $\exists y \in K$, $|x-y| = dist(x, K) = 0 \Longrightarrow x = y \in K \Longrightarrow \bigcap_{\epsilon>0} K_{<\epsilon} \subset K$. Together with the trivial fact that $\bigcap_{\epsilon>0} K_{<\epsilon} \supset K$, we have

$$\bigcap_{\epsilon>0} K_{<\epsilon} = K \Longrightarrow \lim_{\epsilon\to 0^+} \mu(K_{<\epsilon}) = \mu(K)$$

We know that $\forall \epsilon > \epsilon' > 0$, $K_{\leq\epsilon'} \subset K_{\leq\epsilon}$, then

$$\lim_{\epsilon\to 0^+} \mu(K_{\leq\epsilon}) = \mu(\bigcap_{\epsilon>0} K_{\leq\epsilon})$$

We next show that $\bigcap_{\epsilon>0} K_{\leq\epsilon} = K$. $\forall x \in \bigcap_{\epsilon>0} K_{\leq\epsilon}$, $\forall \epsilon > 0$, $dist(x, K) \leq \epsilon \Longrightarrow dist(x, K) = 0$. Since K is a closed set, $\exists y \in K$, $|x-y| = dist(x, K) = 0 \Longrightarrow x = y \in K \Longrightarrow \bigcap_{\epsilon>0} K_{\leq\epsilon} \subset K$. Together with the trivial fact that $\bigcap_{\epsilon>0} K_{\leq\epsilon} \supset K$, we have

$$\bigcap_{\epsilon>0} K_{\leq\epsilon} = K \Longrightarrow \lim_{\epsilon\to 0^+} \mu(K_{\leq\epsilon}) = \mu(K)$$

□

(3)

证明. Note that

$$\bigcap_{\epsilon>0} K_{<\epsilon} = K \Longrightarrow \bigcap_{\epsilon>0} K_{<\epsilon} \setminus K = K \setminus K = \emptyset$$

Hence,

$$\lim_{\epsilon \to 0^+} \mu(K_{<\epsilon} \setminus K) = \mu(\bigcap_{\epsilon > 0} K_{<\epsilon} \setminus K) = \mu(\emptyset) = 0$$

Note that

$$\bigcap_{\epsilon > 0} K_{\leq \epsilon} = K \Longrightarrow \bigcap_{\epsilon > 0} K_{\leq \epsilon} \setminus K = K \setminus K = \emptyset$$

Hence,

$$\lim_{\epsilon \to 0^+} \mu(K_{\leq \epsilon} \setminus K) = \mu(\bigcap_{\epsilon > 0} K_{\leq \epsilon} \setminus K) = \mu(\emptyset) = 0$$

□